Supplemental Material for
"Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes"

Asaf A. Solonnikov and Nadav Katz

The Racah Institute of Physics, The Hebrew University of Jerusalem, Israel

(Dated: July 21, 2026)

This Supplemental Material provides the experimental setup and qubit coherence characterization; a precise definition of the finite echo-root-Lorentzian pulse and its resonant cancellation mechanism; the Bloch-equation coherence limit; pulse-cutoff and adiabaticity analyses; high-resolution and extended-amplitude scans; the numerical model and data-analysis procedure; and comparisons between pulse protocols, simulation, and experiment. The final section gives a three-level AC-Stark calculation that fixes the sign of the anharmonic features and the driven central-resonance displacement, followed by a direct experimental test of resonance-center stability.

S1. MEASUREMENT SETUP

Experiments were performed on a QPU comprising 11 uncoupled superconducting qubits hosted in a dilution refrigerator. Control and readout were implemented with a Quantum Machines OPX1000 equipped with a microwave front-end module (MW-FEM), together with external cryogenic attenuators, filters, circulators, and amplifiers.

A schematic of the setup is shown in Fig. S1. The MW-FEM uses direct digital synthesis and digital upconversion: a programmed complex pulse envelope is translated to the configured microwave carrier and emitted directly from an RF output, without an external analog local oscillator or IQ mixer [1]. For acquisition, the returned microwave signal is digitally downconverted to a complex baseband stream. QUA demodulation then applies cosine and sine integration weights to the sampled signal and accumulates the result to obtain the measured quadratures [2]. Thus, separate MW-FEM outputs provide the qubit-drive and resonator-readout tones, while an MW-FEM input acquires the returning readout signal. A dedicated DC bias line was available but was not used in these measurements; consequently, the qubits were not necessarily operated at their flux sweet spots.

The device was mounted at the mixing-chamber stage ($\sim$9 mK), with the drive and readout lines attenuated across the refrigerator stages. The returning readout signal passed through circulators and a 4 K high-electron-mobility transistor (HEMT) amplifier before entering the MW-FEM for digital downconversion and demodulation.

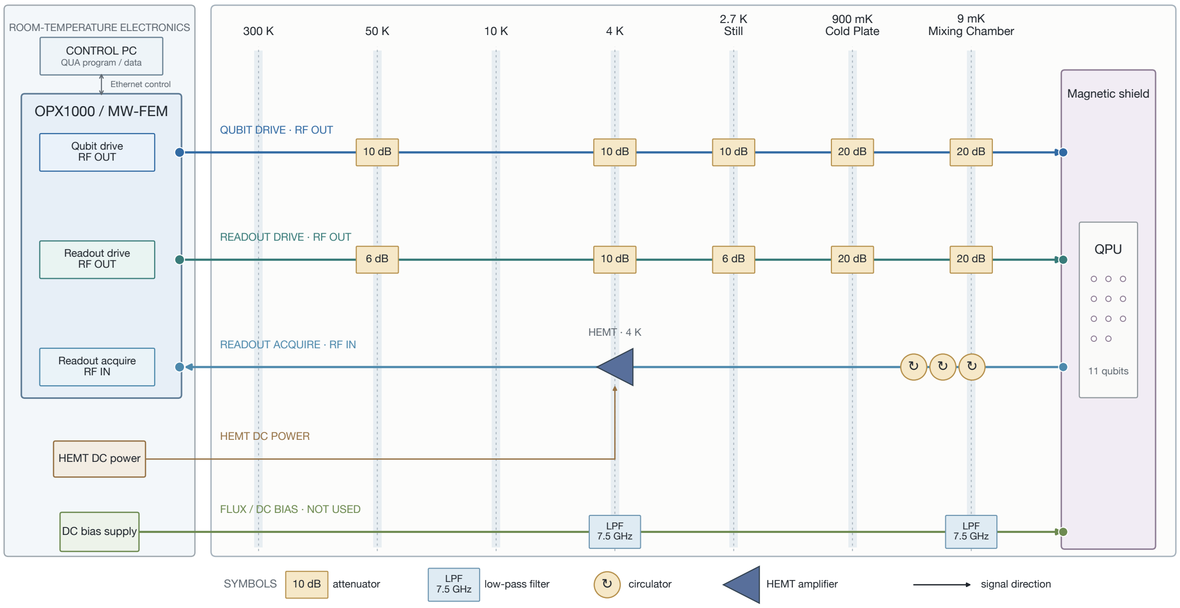

Figure S1. Schematic of the experimental setup for qubit control and readout. A control computer submits QUA programs and receives data over Ethernet. The OPX1000 MW-FEM directly generates the qubit-drive and resonator-readout RF signals and acquires the returning readout signal. Input lines are attenuated across the 300 K, 50 K, 10 K, 4 K, still, cold-plate, and mixing-chamber stages. The low-pass-filtered DC-bias line is shown for completeness but was not used. At the mixing chamber, the device is isolated by circulators; the returning readout signal is amplified by a 4 K HEMT before entering the MW-FEM for digital downconversion and demodulation.

S1.1. Qubit coherence characterization

The energy-relaxation and Ramsey-dephasing times of the qubit used for the spectroscopy measurements were characterized independently. Figure S2 shows representative measurements for qubit q1. Fitting the excited-state population to an exponential decay with a constant offset gives $T_1 = 51.24 \pm 0.73$ μs. Fitting the positive-detuning Ramsey trace to an exponentially damped sinusoid gives $T_2^* = 7.31 \pm 0.13$ μs. The quoted uncertainties are one-standard-deviation estimates obtained from the fit covariance matrices. No associated Hahn-echo trace is available, so no measured Hahn-echo T_2 is reported. When one transverse time is required, we conservatively use $T_{2,\mathrm{eff}} = T_2^* = 7.31$ μs. With the measured T_1, this gives $T_\phi = 7.87$ μs through $T_{2,\mathrm{eff}}^{-1} = (2T_1)^{-1} + T_\phi^{-1}$; this is not Hahn-echo evidence.

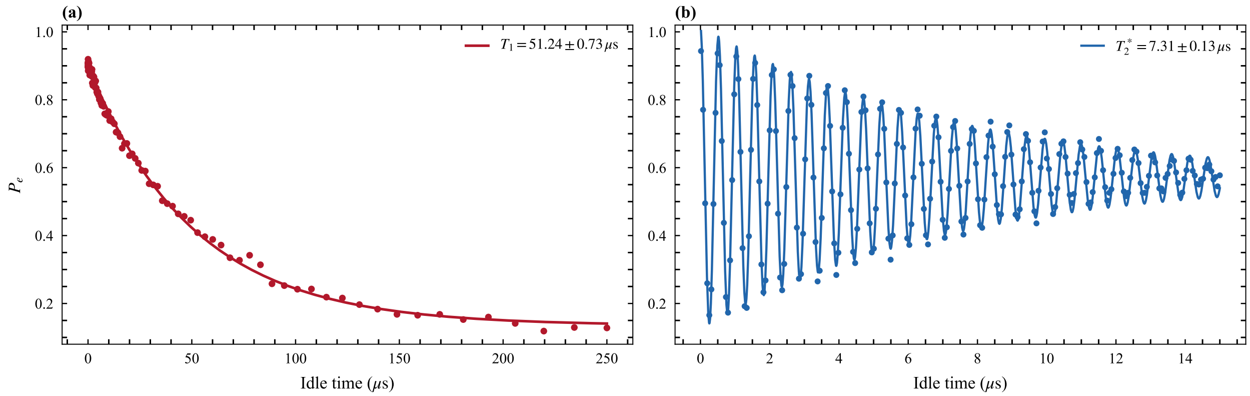

Figure S2. Independent coherence characterization of qubit q1. (a) Energy relaxation following state preparation, together with an exponential fit. (b) Ramsey fringes acquired with a positive 2 MHz programmed detuning, together with an exponentially damped sinusoidal fit.

S2. FINITE PULSE DEFINITION AND ECHO CANCELLATION

We define the finite generalized Lorentzian envelope by

$$f_{n,c,L}(t) = \begin{cases} \left[1 + (t/\tau)^2\right]^{-n}, & |t| \leq L/2, \\ 0, & |t| > L/2, \end{cases} \tag{S1}$$

where L is the total pulse duration, n is the shape exponent, and the relative endpoint amplitude is $c = f_{n,c,L}(L/2)$. These parameters imply

$$\tau = \frac{L/2}{\sqrt{c^{-1/n} - 1}}. \tag{S2}$$

The root-Lorentzian pulse used here corresponds to $n = 1/2$. Its echo version has a phase inversion at the center,

$$\Omega_{\mathrm{E}}(t) = \Omega_0 f_{n,c,L}(t) s(t), \qquad s(t) = \begin{cases} +1, & t < 0, \\ -1, & t \geq 0. \end{cases} \tag{S3}$$

The value assigned at the single point $t = 0$ has no effect in the continuum limit. In a sampled waveform, the two halves are constructed with equal duration and opposite phase.

Two cutoff scans must be distinguished. A pure truncation scan keeps τ, n, and Ω_0 fixed while changing L. A fixed-duration hardware scan instead keeps L fixed and changes the endpoint ratio c; in the parametrization of Eq. (S2), this also changes τ. Results from these two scans need not coincide.

In the rotating-wave approximation, the unitary two-level Hamiltonian is

$$\frac{H(t)}{\hbar} = \frac{1}{2}\left[\Delta\sigma_z + \Omega_{\mathrm{E}}(t)\sigma_x\right]. \tag{S4}$$

At exact resonance, Hamiltonians at different times are proportional to the same operator and therefore commute. The propagator is

$$U(\Delta = 0) = \exp\left[-\frac{i\sigma_x}{2}\int_{-L/2}^{L/2} \Omega_{\mathrm{E}}(t)\,dt\right] = \mathbb{I}, \tag{S5}$$

because the signed areas of the two symmetric halves cancel. For $\Delta \neq 0$, the σ_z and σ_x terms do not commute, so the cancellation is incomplete and a narrow detuning-dependent response remains. Equation (S5) is exact only for a symmetric, unitary two-level pulse. Relaxation, dephasing, phase or amplitude imbalance, and finite waveform bandwidth prevent exact cancellation in the experiment.

S2.1. Numerical illustration of the echo mechanism

Figure S3 shows the numerical qubit-state evolution for the root-Lorentzian pulse and its echo variant. At resonance, the drive produces coherent evolution during the first half of the echo pulse. The π phase inversion reverses this evolution during the second half, so the ideal symmetric sequence returns the qubit to its initial state. In the open-system simulation the return is incomplete because relaxation and dephasing act throughout the pulse, consistent with the limitations of Eq. (S5).

For nonzero detuning, the $\Delta\sigma_z$ term does not reverse at the pulse midpoint and does not commute with the driven evolution. The two pulse halves therefore fail to cancel, leaving a detuning-dependent final population. The contrast between the resonant return and the off-resonant response produces the central depletion feature used to identify the qubit transition.

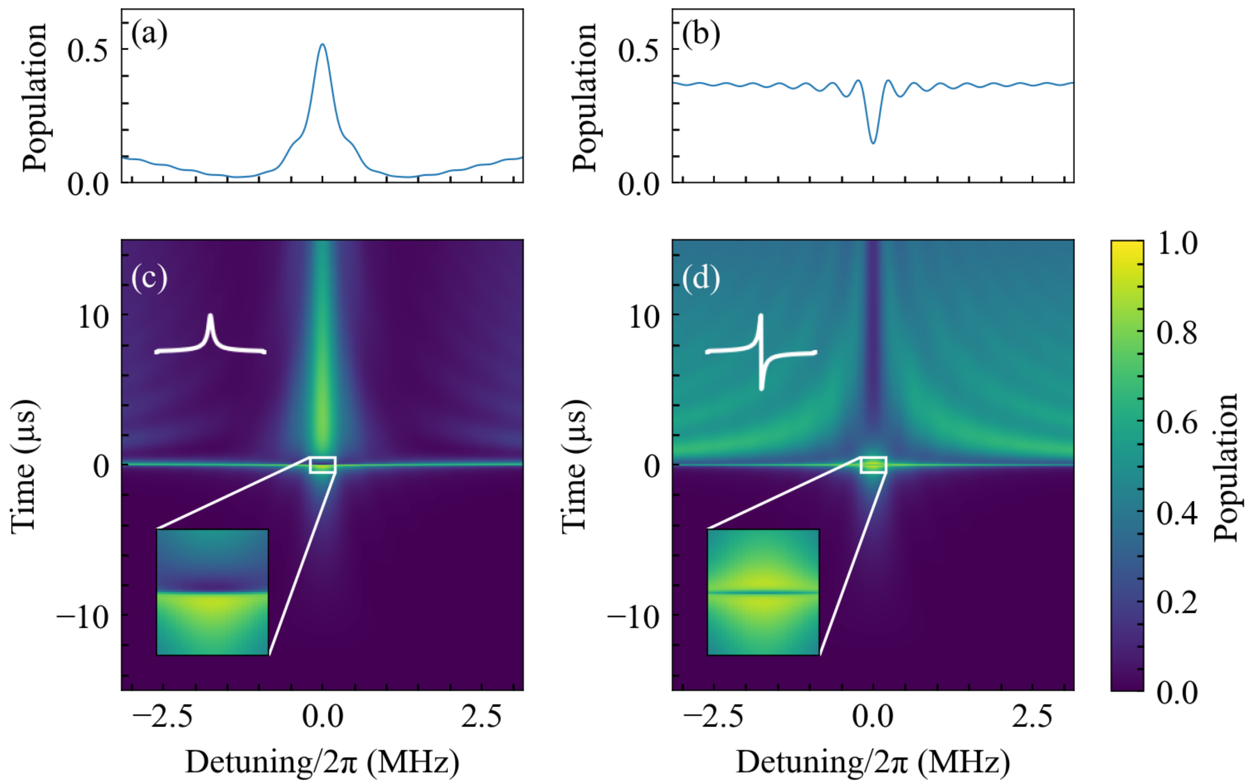

Figure S3. Numerical illustration of the echo-cancellation mechanism. (a) Final excited-state population versus detuning $\Delta/2\pi$ for a root-Lorentzian pulse. (b) Corresponding response for the echo-root-Lorentzian pulse $L^{(1/2)}_{\text{echo}}$. (c,d) Full excited-state evolution versus time and detuning for the two protocols. The pulse duration is 30 μs and the relative endpoint amplitude is 5×10^{-4}. The ordinary root-Lorentzian pulse produces a resonant peak, whereas the phase inversion in the echo pulse reverses the resonant evolution and produces a central dip. At nonzero detuning the two halves do not cancel, leaving the off-resonant population that forms the spectroscopic response. These archived arrays use the historical coherence parameters documented in Sec. S7, rather than the later measurement-linked values.

S3. BLOCH-EQUATION COHERENCE LIMIT

For comparison with the shaped-pulse protocol, we first summarize the steady-state response of a continuously driven two-level system. We define $w \equiv \rho_{00} - \rho_{11}$, with $|0\rangle$ the ground state and $|1\rangle$ the excited state. Hence $P_e = \rho_{11} = (1 - w)/2$; no Pauli matrix sign convention is left implicit. Solving the optical Bloch equations for a drive with angular Rabi frequency Ω_0, angular detuning Δ, relaxation time T_1, and transverse coherence time T_2

gives [3]

$$w_{\rm ss}(\Delta) = \frac{1+\Delta^2 T_2^2}{1+\Delta^2 T_2^2 + \Omega_0^2 T_1 T_2}, \qquad P_{e,\rm ss}(\Delta) = \frac{1}{2}\frac{s}{1+s+\Delta^2 T_2^2}, \quad s \equiv \Omega_0^2 T_1 T_2. \tag{S6}$$

This expression assumes a pulse long enough to approach steady state. It is a reference result for conventional saturation spectroscopy, rather than a closed-form description of the finite echo pulse.

At resonance, $P_{e,\rm ss}(0) = s/[2(1+s)]$, which approaches the saturated value $1/2$ only for $s \gg 1$. The positive half-maximum detuning is the angular-frequency HWHM

$$\Delta_{\rm HWHM} = \frac{1}{T_2}\sqrt{1+s}. \tag{S7}$$

The angular-frequency FWHM is therefore

$$\Gamma_\omega \equiv 2\Delta_{\rm HWHM} = \frac{2}{T_2}\sqrt{1+s}. \tag{S8}$$

The corresponding cyclic-frequency FWHM plotted in the Letter is

$$\Gamma_f \equiv \frac{\Gamma_\omega}{2\pi} = \frac{1}{\pi T_2}\sqrt{1+s}. \tag{S9}$$

Thus $\Delta_{\rm HWHM}$, Γ_ω, and Γ_f are distinct quantities and are not denoted by a common symbol.

In the weak-drive limit $s \ll 1$,

$$\Gamma_{\omega,T_2} = \frac{2}{T_2}, \qquad \Gamma_{f,T_2} = \frac{1}{\pi T_2}, \tag{S10}$$

which are the same coherence-limited FWHM expressed in angular frequency and hertz, respectively. In the strong-drive limit $s \gg 1$,

$$\Gamma_\omega \longrightarrow 2\Omega_0\sqrt{\frac{T_1}{T_2}}, \qquad \Gamma_f \longrightarrow 2f_R\sqrt{\frac{T_1}{T_2}}, \tag{S11}$$

where $f_R = \Omega_0/(2\pi)$. These limits are governed by s, not by comparing Ω_0 with an ambiguously defined linewidth.

S4. PULSE TRUNCATION AND ENDPOINT AMPLITUDE

The ideal Lorentzian tail has infinite support, whereas the implemented pulse starts and ends at finite amplitude $\Omega_c = c\Omega_0$. Abruptly switching this residual drive produces an endpoint transient. A useful estimate of the associated population contribution is [4]

$$P_c(\Delta) \simeq \frac{\Omega_c^2}{\Omega_c^2+\Delta^2}\left[1-P_0(\Delta)\right], \qquad \Omega_c = c\Omega_0, \tag{S12}$$

where P_0 denotes the response that would be obtained without the endpoint jump. The artifact becomes important when Ω_c is comparable to the linewidth scale of interest. Thus the cutoff cannot be assessed from c alone: it must be considered together with the peak drive and the desired resolution.

The Bloch-equation reference gives the dimensionless endpoint saturation parameter

$$s_c = \Omega_c^2 T_1 T_2. \tag{S13}$$

Keeping the endpoint contribution in the weak-saturation regime requires

$$s_c \ll 1 \quad \Longleftrightarrow \quad \Omega_c \ll \frac{1}{\sqrt{T_1 T_2}} \quad \Longleftrightarrow \quad c \ll \frac{1}{\Omega_0 \sqrt{T_1 T_2}}. \tag{S14}$$

The inequality therefore bounds the endpoint amplitude from above. It is a perturbative design criterion, not a claim that the cutoff artifact has the steady-state Lorentzian lineshape.

We therefore measured the cutoff dependence over two complementary q1 echo–root–Lorentzian campaigns. The broad survey spans $c = 0.002$–0.99 with 25 peak-amplitude settings, while the focused sweep resolves $c = 0.01$–0.10 with 200 peak-amplitude settings. Both use a 20 μs pulse and the same measured $T_2 = 6.856$ μs. Figure S4 reports the dimensionless reciprocal-linewidth metric

$$\mathcal{R} = \frac{\Gamma_{f,T_2}}{\Gamma_f} = \frac{1/(\pi T_2)}{\Gamma_f}, \tag{S15}$$

where every linewidth is a cyclic-frequency FWHM in hertz. Thus $\mathcal{R} = 1$ is the T_2-limited linewidth and $\mathcal{R} < 1$ denotes a broader feature. We also show the signal-weighted metric $|A_{\text{fit}}|\mathcal{R}$, which suppresses narrow fits with negligible contrast. These are linewidth-based scores, not direct estimates of frequency-estimation uncertainty.

The maps retain only resolved and bounded probability features: the fitted contrast is between 0.05 and 1, $R^2 \geq 0.60$, the fitted center lies inside the measured frequency interval, and the FWHM lies between two frequency bins and half the sweep span. Gray cells denote measurements for which the fit did not pass this common screen. The denser focused sweep provides substantially more accepted fits than the coarse survey and resolves the loss of reciprocal linewidth performance as the cutoff increases across the 0.01–0.10 interval.

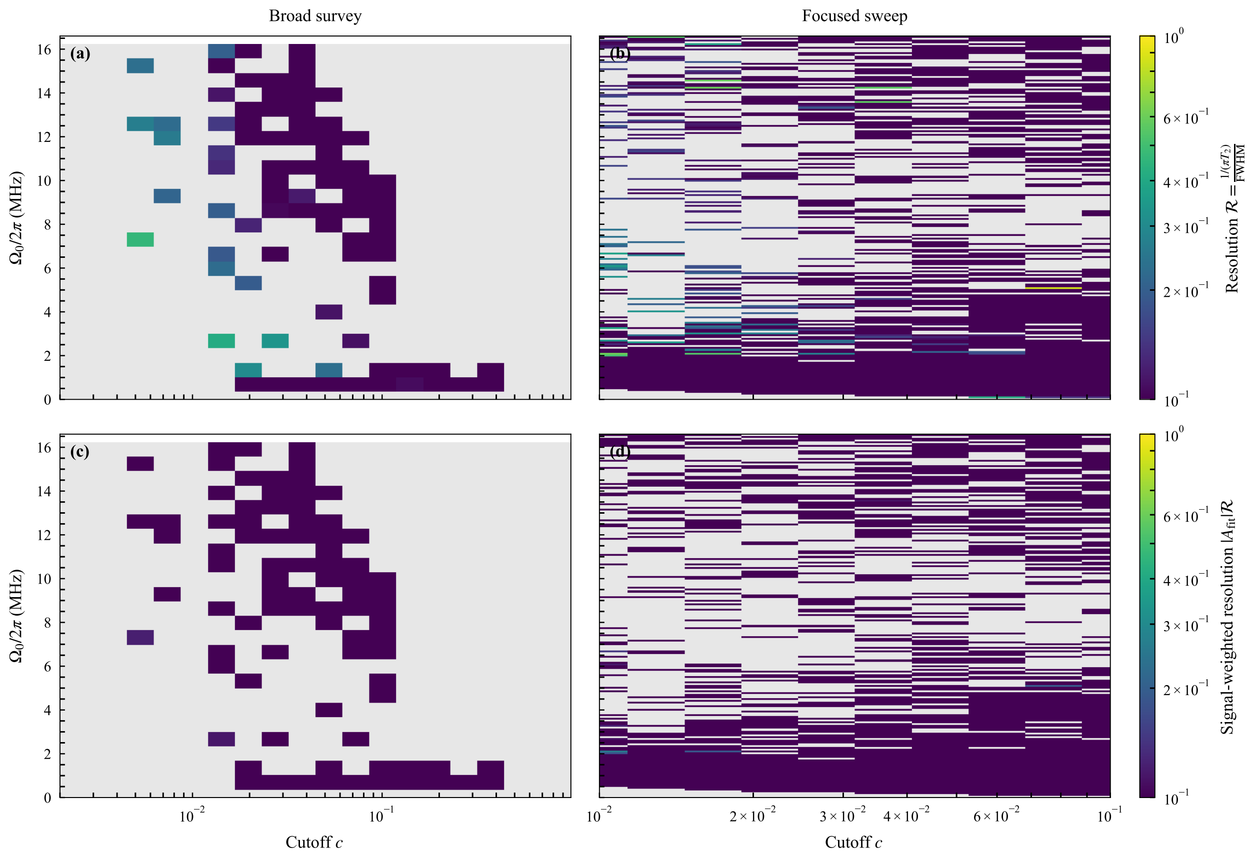

Figure S4. Experimental cutoff sweep for q1 echo–root–Lorentzian spectroscopy. The left column is the broad `cutoff_amp_fwhm_map` campaign ($c = 0.002$–0.99, 25 amplitudes, 60 repetitions per point); the right column is the denser `echo_lorentzian_cutoff_sweep` campaign ($c = 0.01$–0.10, 200 amplitudes, 100 repetitions per point). (a,b) Reciprocal-linewidth metric $\mathcal{R}$ and (c,d) signal-weighted metric $|A_{\rm fit}|\mathcal{R}$. Both color scales are logarithmic. On the upper color scale, $\mathcal{R} = 1$ is the T_2-limited linewidth and values below one correspond to FWHM broader than that limit. Gray cells are rejected or unresolved fits under the common screen described in the text; no fitted surface or interpolation is shown.

S4.1. Simulated cutoff–amplitude operating map

The finite-duration pulse is specified by its duration, peak drive amplitude, and cutoff level. To isolate the latter two parameters, we fix the pulse duration at 50 μs and simulate the root-Lorentzian and echo-root-Lorentzian protocols over matched amplitude–cutoff grids. For each simulated spectrum, we extract the cyclic-frequency FWHM Γ_f and the contrast-weighted width $\Gamma_f/(A_{\rm peak}\Gamma_{f,T_2})$. This is a heuristic resolution–signal metric; a direct Fisherinformation or Cramér–Rao analysis would additionally require a measurement-noise model and the full frequency dependence of the response.

Figure S5 shows that the root-Lorentzian pulse contains extended narrow-linewidth regions, but its useful operating structure is strongly oscillatory. The echo protocol instead produces a broader, contiguous region in which both the linewidth and the resolution–signal trade-off remain favorable. The logarithmic parameter axes should be kept in mind when comparing the apparent area of these regions.

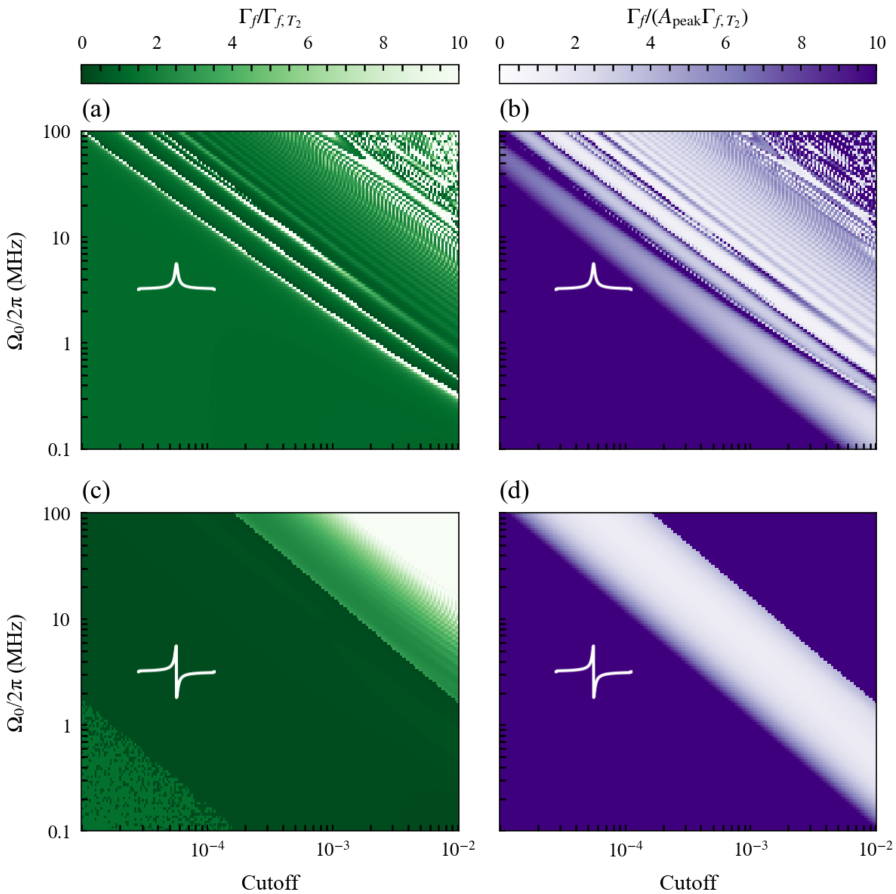

Figure S5. Numerical cutoff–amplitude maps for root-Lorentzian $L^{(1/2)}$ and echo-root-Lorentzian $L^{(1/2)}_{\text{echo}}$ pulses of duration 50 μs. (a,c) Normalized cyclic-frequency FWHM $\Gamma_f/\Gamma_{f,T_2}$ for the root and echo protocols, respectively; (b,d) corresponding contrast-weighted widths $\Gamma_f/(A_{\text{peak}}\Gamma_{f,T_2})$. The ordinary root-Lorentzian response is strongly oscillatory, whereas the echo variant retains a broader contiguous region with a favorable resolution–signal trade-off.

S5. ADIABATIC-BASIS DESCRIPTION

For one half of the pulse, the rotating-frame Hamiltonian can be written

$$\frac{H(t)}{\hbar} = \frac{1}{2}\begin{bmatrix} \Delta & \Omega(t) \\ \Omega(t) & -\Delta \end{bmatrix}. \tag{S16}$$

Define

$$\epsilon(t) = \sqrt{\Delta^2 + \Omega^2(t)}, \qquad \vartheta(t) = \frac{1}{2}\,\mathrm{atan2}[\Omega(t), \Delta]. \tag{S17}$$

Both ϵ and $\dot{\vartheta}$ have units of inverse seconds, whereas ϑ is dimensionless. The instantaneous eigenvalues are $E_- = -\hbar\epsilon/2$ and $E_+ = +\hbar\epsilon/2$. Taking the instantaneous eigenvectors as the columns of

$$R(t) = \begin{bmatrix} -\sin\vartheta & \cos\vartheta \\ \cos\vartheta & \sin\vartheta \end{bmatrix}, \tag{S18}$$

the transformed Hamiltonian is $H_a = R^{\mathsf{T}} H R - i\hbar R^{\mathsf{T}} \dot{R}$. Therefore

$$\frac{H_a(t)}{\hbar} = \begin{bmatrix} -\epsilon/2 & -i\dot{\vartheta} \\ i\dot{\vartheta} & \epsilon/2 \end{bmatrix}, \qquad \dot{\vartheta} = \frac{\Delta\dot{\Omega}}{2(\Delta^2 + \Omega^2)}. \tag{S19}$$

The nonadiabatic coupling is $|\dot{\vartheta}|$, while the instantaneous angular-frequency gap is ϵ. Adiabatic following therefore requires [5]

$$|\dot{\vartheta}| \ll \epsilon. \tag{S20}$$

Replacing the inequality by equality provides the local crossover estimate

$$\sqrt{\Delta_b^2 + \Omega^2(t_0)} = \frac{|\Delta_b \dot{\Omega}(t_0)|}{2[\Delta_b^2 + \Omega^2(t_0)]}, \tag{S21}$$

where t_0 is the time at which the nonadiabatic coupling is largest for the detuning under consideration. This is a scaling criterion, not an exact linewidth formula.

This smooth-basis construction applies separately to each half of the echo pulse; the imposed phase jump at $t = 0$ is a control discontinuity and is not represented by $\dot{\vartheta}$ in this local adiabatic estimate.

For $L^{(n)}(t) \sim |t|^{-2n}$, the inverse-power-tail derivation of Boradjiev and Vitanov requires $2n > 1$. For an untruncated generalized Lorentzian with $n > 1/2$, it gives

$$\Delta_b \tau \propto (\Omega_0 \tau)^{-1/(2n-1)}. \tag{S22}$$

As $n \to 1/2^+$, the exponent $1/(2n-1)$ diverges, indicating the strongest power-narrowing trend within the asymptotic family. The exact root-Lorentzian case $n = 1/2$ is nevertheless marginal and lies outside the domain of Eq. (S22); substituting $n = 1/2$ into that power law is invalid. In particular, the singularity does not imply zero FWHM. The finite pulse length, nonzero endpoint amplitude, decoherence, and echo phase inversion set the observed Γ_f, which is evaluated from the complete simulated and experimental response in this work.

S6. LARGE-DRIVE SPECTROSCOPY

S6.1. Broad and Narrow Frequency Scans

High-resolution spectroscopy at large drive amplitudes is characterized on two frequency scales. A broad scan first locates the qubit transition and reveals the surrounding baseline, nearby spectral features, and any strong-drive distortions. A narrow scan centered on the transition then samples the sharp echo–root–Lorentzian depletion feature densely enough to determine its center and linewidth. Presenting the broad and narrow scans together distinguishes a genuinely narrow resonance from a local feature embedded in a wider distorted response and demonstrates that the high resolution persists within the full spectral landscape.

S6.2. Extended-Amplitude Regime and Strong-Drive Distortions

At drive amplitudes above approximately 50 MHz, the Rabi frequency is no longer small compared with the transmon anharmonicity and the measured response begins to depart from an effective two-level description. Possible mechanisms include coupling to higher transmon levels, drive-induced level shifts, leakage, and multiphoton processes. The two-level calculation in Sec. S7 cannot distinguish among these mechanisms, so the additional structure is reported here as a strong-drive distortion rather than assigned to a specific transition. Quantitative attribution would require a converged multilevel model and an independent calibration of the delivered drive amplitude.

S6.3. Cutoff Dependence from 50 to 80 MHz

To separate large-drive behavior from changes in pulse duration or sampling, we compare three q1 echo–root–Lorentzian scans acquired with the same 10 μs duration, detuning grid, amplitude grid, active-reset protocol, and 40 repetitions per point. Only the dimensionless cutoff-control setting is changed, from $c = 0.001$ to 0.005 and 0.010. Using the same-day q1 π-pulse calibration retained with the data, the displayed portion of each scan is selected from raw peak Rabi frequencies of 50.28 to 76.97 MHz. The centers of the displayed block averages span 50.67 to 76.19 MHz. This restricted view exposes the high-amplitude response directly instead of allowing the much larger low-amplitude region to dominate the color scale. To reduce binomial sampling noise without imposing a fit or interpolated surface, each displayed cell is a non-overlapping average of three adjacent detuning points and three adjacent amplitude points. Since every raw point contains 40 repetitions, a displayed cell averages 360 binary outcomes.

Figure S6 shows a pronounced reorganization of the large-drive response with cutoff. At $c = 0.001$ the map is dominated by a step-like change across zero detuning, whereas the $c = 0.005$ and 0.010 scans develop curved, amplitude-dependent fringes that cross the central-frequency region. Thus these measurements do not support extrapolating the low-amplitude narrow-feature behavior unchanged into the 50–80 MHz regime. Instead, they identify a cutoff-sensitive strong-drive regime in which a single-linewidth description is insufficient. Because all panels use one color scale, the differences are not artifacts of independent normalization.

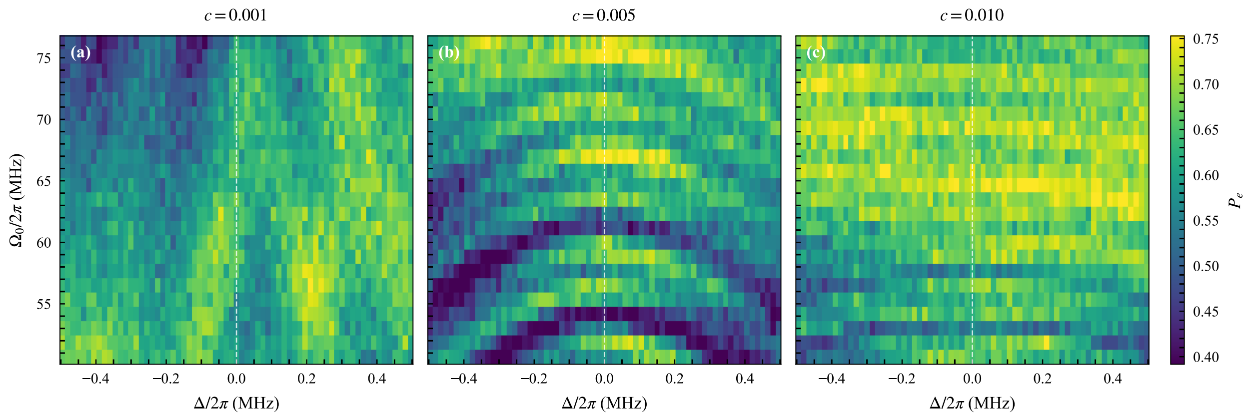

Figure S6. Measured high-amplitude echo–root–Lorentzian spectroscopy on q1 for dimensionless cutoff-control settings (a) $c = 0.001$, (b) $c = 0.005$, and (c) $c = 0.010$. Each panel shows the same 50.28–76.97 MHz raw peak-Rabi-frequency interval from a 10 μs scan with 40 repetitions per point. The dashed line marks zero detuning, and all panels share the color scale. Displayed cells are non-overlapping 3×3 block averages of the measured grid; no interpolation or fitted surface is shown. The progression from a step-like response in (a) to curved interference fringes in (b) and (c) demonstrates that the large-drive structure is cutoff sensitive.

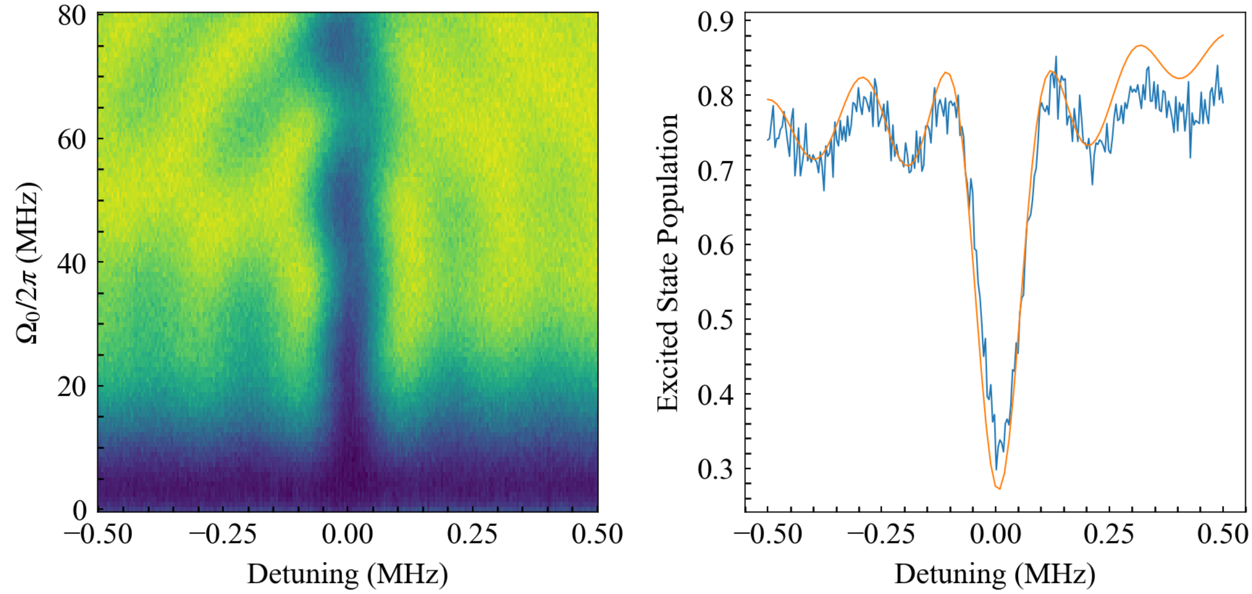

Figure S7. Two-dimensional spectroscopy map obtained with a 10 μs echo–root–Lorentzian pulse at extended drive amplitudes (up to 80 MHz). The left panel shows the full detuning–amplitude map, while the right panel presents a line cut at fixed drive strength (blue) together with a smooth fit (orange). The observed distortions mark the breakdown of the ideal effective two-level response as the drive approaches the multilevel regime.

S7. NUMERICAL MODEL

Unless otherwise noted, the simulations use an effective driven two-level system coupled to Markovian relaxation and dephasing channels. Main-text Fig. 2 instead uses the three-state transmon extension described below. In the frame rotating at the drive frequency, and within the rotating-wave approximation, the Hamiltonian is implemented as

$$\frac{H(t)}{\hbar} = -\Delta a^\dagger a + \frac{\alpha}{2} a^{\dagger 2} a^2 + \frac{\Omega_0}{2} f(t) \left(a + a^\dagger\right), \tag{S23}$$

where $\Delta = \omega_d - \omega_{01}$ is the drive–qubit detuning, Ω_0 is the peak Rabi frequency, α is the anharmonicity, and a is the truncated oscillator lowering operator. In the effective two-level model, $a = |0\rangle\langle 1|$ and the anharmonicity term vanishes. Up to an irrelevant identity term, Eq. (S23) is equivalent to $H/\hbar = (\Delta/2)\sigma_z + (\Omega_0 f/2)\sigma_x$.

The simulations use the finite pulse in Eqs. (S1)–(S3). The overall sign of the echo envelope is a phase convention and does not affect the measured populations.

The density matrix evolves according to the Lindblad master equation

$$\dot{\rho} = -\frac{i}{\hbar}[H(t), \rho] + \mathcal{D}\left[\sqrt{\frac{1}{T_1}}\, a\right]\rho + \mathcal{D}\left[\sqrt{\frac{2}{T_\phi}}\, a^\dagger a\right]\rho, \tag{S24}$$

where $\mathcal{D}[C]\rho = C\rho C^\dagger - \frac{1}{2}\{C^\dagger C, \rho\}$ and

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}. \tag{S25}$$

The coherence characterization used by measurement-linked calculations is given in the measurement-setup section. The archived figure arrays were generated with the historical inputs $T_1 = 30$ μs and $T_\phi = 20$ μs, which imply $T_2 = 12$ μs. These values describe the cached numerical data, not the later q1 coherence measurement. A measurement-linked calculation instead uses $T_1 = 51.24 \pm 0.73$ μs and, in the absence of a Hahn-echo measurement, the conservative choice $T_{2,\mathrm{eff}} = T_2^* = 7.31$ μs, corresponding to $T_\phi = 7.87$ μs. A measured Hahn-echo T_2 should replace this effective choice when such a trace is acquired. Each calculation starts in $|0\rangle$, integrates Eq. (S24) over the complete pulse, and records the final state observables. Spectroscopy maps are generated by repeating this evolution over the experimental detuning and drive-amplitude grids, using the pulse duration, cutoff, T_1, and T_2 associated with the corresponding data set. The time-dependent master equation is solved with QuTiP or, for the dense matched sweep in Figs. S8–S10, the algebraically equivalent two-level Bloch equations with fourth-order Runge–Kutta integration. The dense main-text Fig. 2 maps use the states $|g\rangle$, $|e\rangle$, and $|f\rangle$, with $\alpha/(2\pi) = -200$ MHz, and report $P_e = \rho_{ee}$ while retaining $P_f = \rho_{ff}$ in the archived figure data. Other production calculations use a two-dimensional Hilbert space and a uniform temporal sampling of the complete waveform. The default solver grid contains 1000 points; the matched linewidth sweep uses 2000 steps. The even grids contain equal numbers of negative- and positive-time samples, with the central phase inversion between them.

Except for main-text Fig. 2, the numerical Hilbert space remains restricted to two levels. Those two-level calculations capture coherent drive dynamics, relaxation, and pure dephasing, but do not describe leakage, higher-level AC-Stark shifts, or multiphoton transitions. Comparisons based on them in the extended-amplitude regime are therefore interpreted as tests of where the effective description breaks down rather than as a simulation of those multilevel effects.

S8. EXPERIMENTAL–SIMULATION COMPARISON

The amplitude-robustness data set was acquired on qubit q1 using active reset and state-discriminated readout. Each point contains 40 repetitions. The detuning axis spans -0.5 to $+0.5$ MHz in 1 kHz steps, and the data comprise 100 fixed-amplitude slices. The pulse record contains a root–Lorentzian edge-to-peak ratio $c = 0.005$, a pulse length $L = 10$ μs, and an unscaled peak output amplitude of 0.2. The peak Rabi frequency is reconstructed from the saved q1 square-pulse calibration, including this peak output amplitude; the resulting experimental range is 0 to 15.32 MHz.

For the matched comparisons in Figs. S8–S10, we integrate the two-level Lindblad model of Eq. (S24) on the complete experimental detuning and calibrated peak-Rabi grids. The calculation uses the recorded echo–root–Lorentzian pulse parameters $L = 10$ μs and $c = 0.005$ together with the measurement-linked values $T_1 = 51.24 \pm 0.73$ μs and $T_\phi = 7.87$ μs. No affine readout calibration or fitted frequency offset is applied.

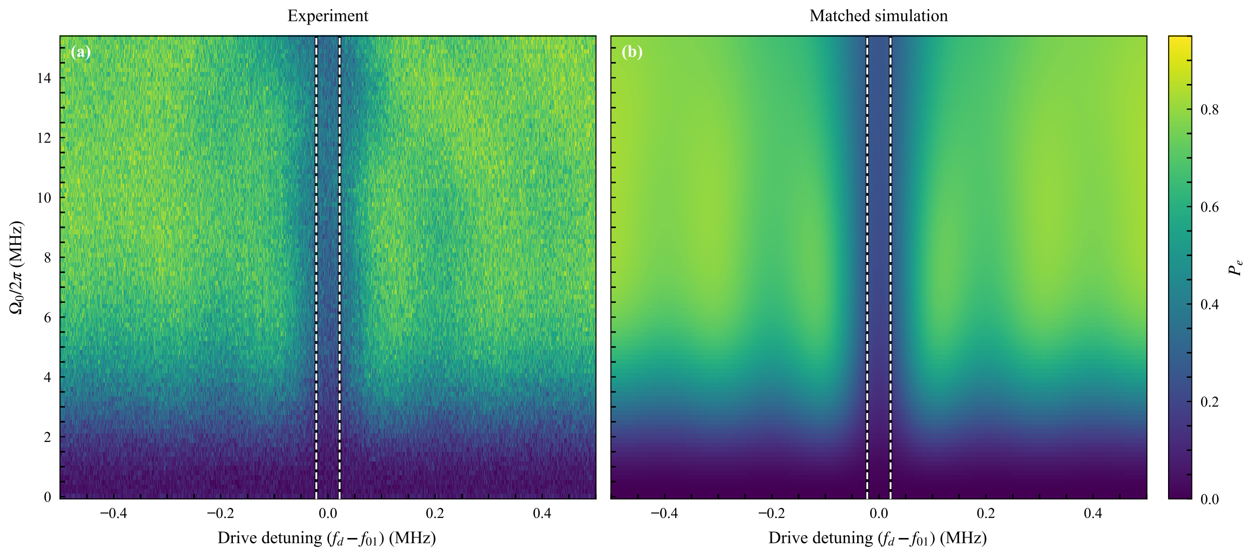

Figure S8. Complete amplitude–detuning sweeps for the q1 echo–root–Lorentzian sequence. (a) Measured excited-state probability from 40 repetitions per point. (b) Parameter-matched two-level Lindblad simulation on the identical calibrated peak-Rabi and detuning grids. The two panels use the same probability color scale. Both show the narrow resonance-centered cancellation channel, while the deterministic model is smoother and predicts a higher off-resonant excitation probability at intermediate and high drive. The paired black-edged white dashed lines at $\delta = \Delta/(2\pi) = \pm\Gamma_{f,T_2}/2 = \pm 21.75$ kHz enclose the conservative cyclic-frequency T_2-limited FWHM, $\Gamma_{f,T_2} = 43.5$ kHz.

Figure S9 resolves this comparison at three representative amplitudes spanning the accepted operating range. The central dip and its drive-dependent broadening are reproduced by the model. The remaining disagreement is primarily in the off-resonant background and side-lobe structure, and therefore appears most directly in the extracted contrast rather than in the central linewidth.

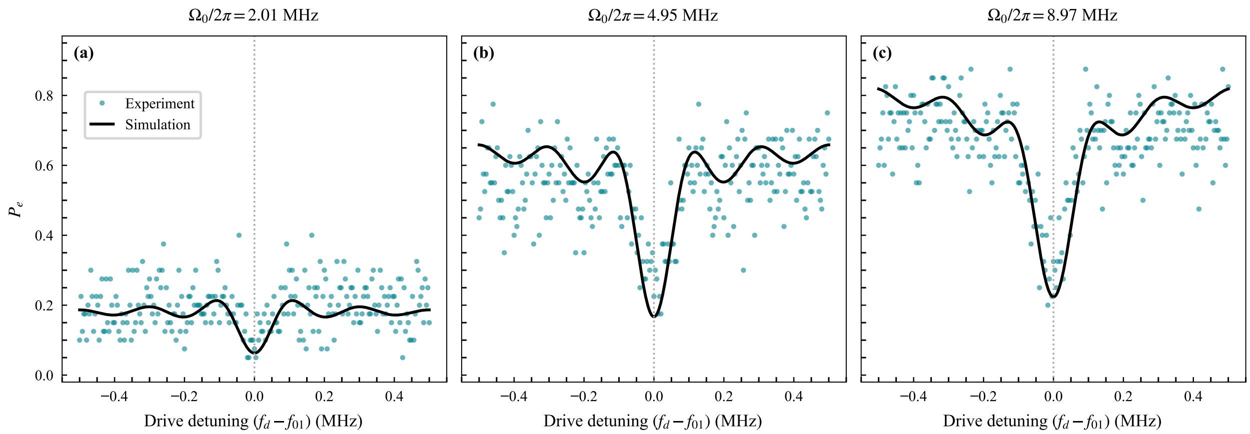

Figure S9. Measured (colored markers) and simulated (black curves) fixed-amplitude spectra at calibrated peak Rabi frequencies $\Omega_0/2\pi = 2.011$, 4.951, and 8.973 MHz. The vertical dotted line denotes zero detuning. For visual clarity, every fourth measured detuning point is displayed; all 1001 points enter the linewidth analysis. The simulation uses the parameters of Fig. S8 without a readout rescaling or fitted frequency shift.

To quantify the comparison across the complete sweep, each fixed-amplitude trace is analyzed independently. We discard 40 points from each frequency edge and apply a Gaussian filter with a standard deviation of one sample to suppress single-bin noise. The remaining data are fit to the operational dip estimator

$$y(\delta) = y_0 - A\exp\left[-\frac{(\delta-\delta_0)^2}{2\sigma_f^2}\right], \tag{S26}$$

where $\delta = \Delta/(2\pi)$ is cyclic detuning, A is the contrast, δ_0 is the fitted center, and

$$\Gamma_f = 2\sqrt{2\ln 2}\,|\sigma_f|. \tag{S27}$$

The Gaussian is an operational, consistent linewidth estimator; it is not a claim that the complete physical response is exactly Gaussian.

We normalize the fitted linewidth to the conservative measurement-linked coherence reference

$$\Gamma_{f,T_2} = \frac{1}{\pi T_{2,\text{eff}}} = 43.5\text{ kHz}, \qquad T_{2,\text{eff}} = 7.31\ \mu\text{s}. \tag{S28}$$

The corresponding dimensionless linewidth, normalized resolution, and contrast-weighted resolution are

$$W = \frac{\Gamma_f}{\Gamma_{f,T_2}}, \qquad R = \frac{\Gamma_{f,T_2}}{\Gamma_f}, \qquad Q = AR. \tag{S29}$$

Thus, $W = 1$ and $R = 1$ denote the coherence-reference linewidth and its reciprocal resolution, respectively. The metric Q additionally penalizes a narrow feature when its measured dip contrast is small.

The simulated traces are smoothed, fit, normalized, and quality-gated by exactly the same procedure as the experimental traces. Because this is a deterministic model curve, its local least-squares uncertainties are not displayed as sampling-error bars.

Fits are accepted only when all parameters and covariance-derived uncertainties are finite, $A \geq 0.02$, $\Gamma_f \leq 0.12$ MHz, and $|\delta_0| \leq 0.050$ MHz. The reported parameter uncertainties are one standard deviation from the fit covariance matrix. Of the 100 amplitude slices, 62 pass these numerical gates. Thirty are rejected as too broad, seven have centers outside the accepted window, and one fails both the contrast and center gates. Rejected or failed fits remain missing values; they are never replaced by zero linewidth or zero uncertainty. Of the 100 matched simulation slices, 54 pass the same gates; 41 are too broad, four have contrast below threshold, and the zero-drive slice fails both the contrast and center gates.

For direct visual comparison with the constant-drive reference below, we use the same operational threshold $Q \geq 0.10$. On the sampled matched-simulation grid this selects $\Omega_0/2\pi = 2.475$–8.973 MHz; the maximum simulated value, $Q = 0.219$, occurs at 6.034 MHz.

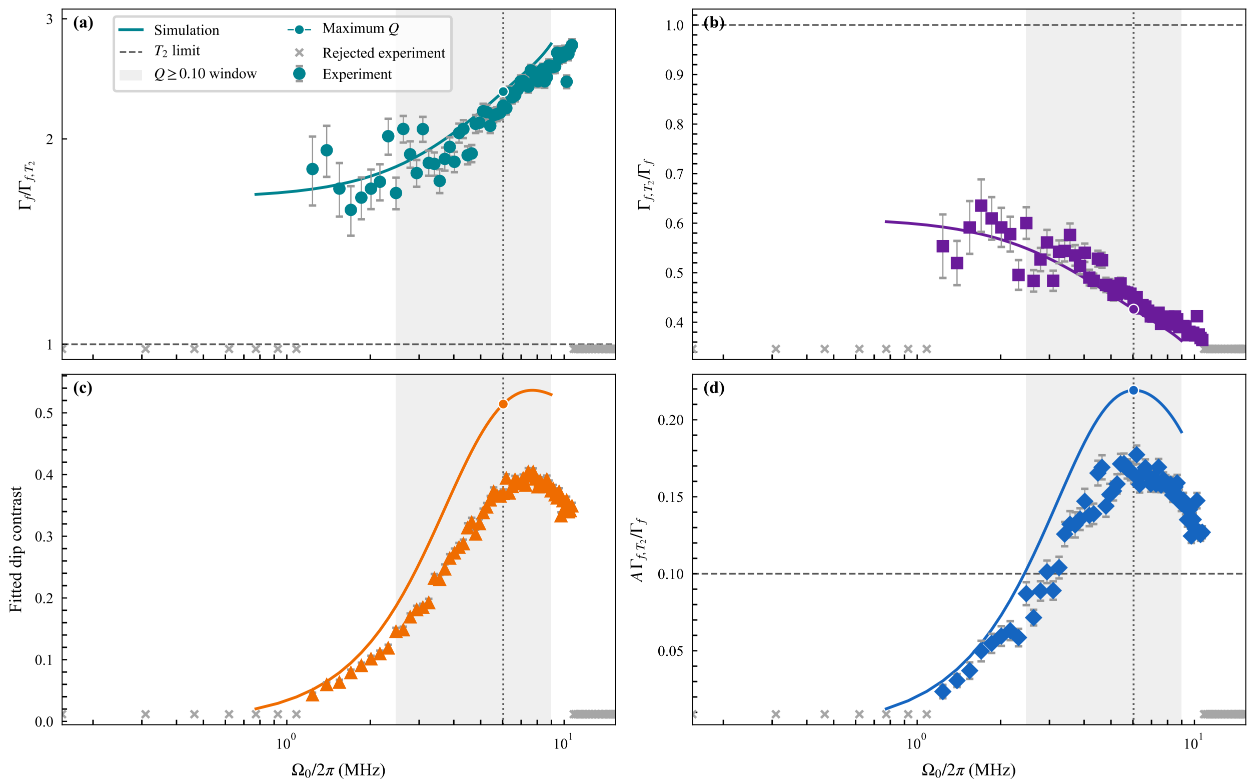

Figure S10. Amplitude dependence of the measured q1 echo–root–Lorentzian response (colored markers) and the parameter-matched two-level Lindblad simulation (same-colored curves). The experimental markers are circles, squares, triangles, and diamonds in panels (a)–(d), respectively. (a) Fitted FWHM in units of the conservative coherence reference Γ_{f,T_2}. (b) Normalized resolution $R = \Gamma_{f,T_2}/\Gamma_f$. (c) Fitted dip contrast A. (d) Contrast-weighted resolution $Q = AR$. Experimental vertical error bars are propagated from the fit covariance matrix; panel (d) includes the fitted contrast–width covariance. Gray crosses along the lower edge mark experimental amplitude slices rejected by the stated quality gates and are not plotted as zero-valued measurements. Simulation slices that fail the same gates appear as gaps in the corresponding colored curves. As in Fig. S11, the gray band marks the matched-simulation interval with $Q \geq 0.10$; the dotted vertical line and circular curve markers identify the maximum simulated Q. The dashed horizontal line in panel (d) is the same $Q = 0.10$ operational threshold.

S8.1. Constant-pulse reference window

For comparison, we use the analytic steady-state Torrey constant-drive tool employed by the IPS FWHM plots. With saturation parameter $s = \Omega_0^2 T_1 T_2$, the power-broadened linewidth and population contrast are

$$W \equiv \frac{\Gamma_f}{\Gamma_{f,T_2}} = \sqrt{1+s}, \qquad A = \frac{s}{2(1+s)}. \tag{S30}$$

Here $T_1 = 51.24 \pm 0.73$ μs and $T_2 = 7.31$ μs, consistent with the normalization used above. We also report the normalized resolution $R = 1/W$ and the contrast-weighted resolution $Q = AR$ over the displayed constant-drive range $\Omega_0/2\pi = 10^{-3}$–10^0 MHz. A logarithmic amplitude axis resolves the low-drive trade-off while retaining the power-broadened regime.

Figure S11 exposes the expected trade-off: contrast rises toward its saturation value while power broadening reduces resolution. In the low-drive limit, $W \to 1$, so the FWHM approaches the T_2-limited value $\Gamma_{f,T_2} = 1/(\pi T_2) = 43.5$ kHz. Using $Q \geq 0.10$ as an operational visualization threshold gives one interval, $\Omega_0/2\pi = 0.0045$–0.0385 MHz. The analytic maximum occurs at $\Omega_0/2\pi = 0.0116$ MHz, where $W = 1.732$, $R = 0.577$, $A = 0.333$, and $Q = 0.192$. The threshold is used only to highlight the small amplitude window that preserves both signal and resolution; it is not a universal physical boundary.

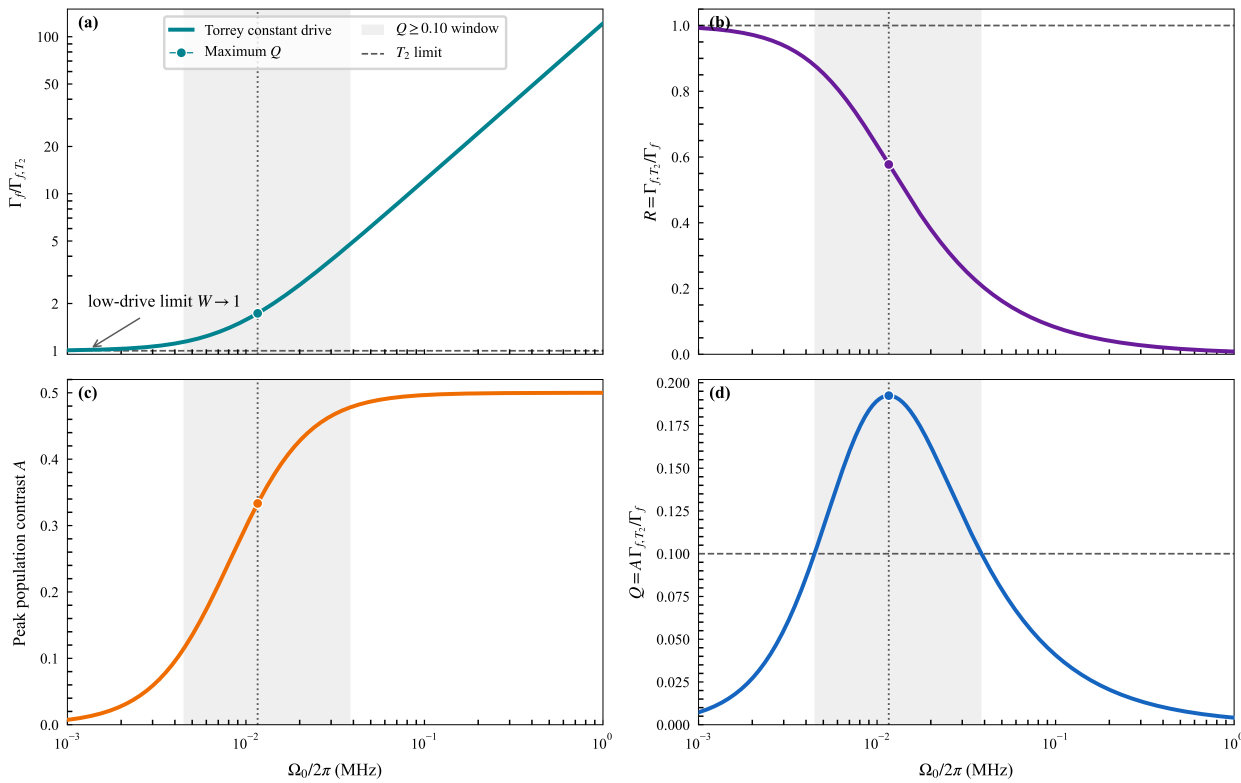

Figure S11. Analytic steady-state Torrey metrics for a constant drive. (a) FWHM in units of Γ_{f,T_2}; the curve explicitly approaches $W = 1$ in the low-drive limit. (b) Normalized resolution $R = \Gamma_{f,T_2}/\Gamma_f$. (c) Peak population contrast A. (d) Contrast-weighted resolution $Q = AR$. The shaded vertical band marks the interval $Q \geq 0.10$, and the dotted vertical line and circular markers identify the maximum of Q. Dashed horizontal lines indicate the T_2 reference in panels (a) and (b) and the operational Q threshold in panel (d). The constant-drive amplitude is shown on a logarithmic axis.

S9. ROOT-LORENTZIAN VERSUS ECHO-ROOT-LORENTZIAN PULSES

The two protocols are compared over the principal experimental control parameters: pulse duration, relative cutoff, and peak drive amplitude. For each parameter set, we examine the final spectroscopic response, extracted linewidth, signal contrast, and linewidth-to-signal ratio. Pulse-length sweeps distinguish the Fourier-limited and decoherence-limited regimes, while cutoff sweeps quantify broadening introduced by truncating the formally infinite Lorentzian tail. Drive-amplitude sweeps then test whether the narrow feature remains stable as the Rabi frequency is increased. These comparisons show where the phase inversion of the echo-root-Lorentzian pulse suppresses coherent oscillations and produces a broader, more uniform operating region than the corresponding root-Lorentzian pulse.

S9.1. Experimental Long-Pulse Comparison

At fixed cutoff and peak Rabi frequency, varying the pulse length separates short-pulse Fourier broadening from the long-pulse limit imposed by decoherence. The comparison also reveals amplitude-dependent coherent oscillations in the root-Lorentzian response and their suppression by the echo phase inversion.

The OPX1000 data archive contains a controlled pulse-duration series acquired on qubit q9 with matched root-Lorentzian and echo-root-Lorentzian scans spanning $L = 2$ to 250 μs. Figure S12 shows three representative durations across the series.

The complete measured series contains $L = 2$, 5, 10, 30, 60, 90, 130, 160, and 250 μs. All six scans displayed in Fig. S12 use the same -2.5 to $+2.5$ MHz detuning grid, amplitude-prefactor range, 100 repetitions per point, relative cutoff $c = 5 \times 10^{-4}$, and peak waveform amplitude. For $L = 160$ μs, the OPX1000 plays a 60 μs stored template stretched to the recorded physical duration. This preserves the normalized envelope while avoiding an unnecessarily long arbitrary waveform record.

The convergence at long duration has a simple physical interpretation. The pulse is long enough for the adiabatic passage to be completed for either phase convention, while T_1 relaxation during the drive washes out the protocol-dependent coherent population structure away from resonance. The broad off-resonant region is therefore driven toward the same saturated background for the root-Lorentzian and echo-root-Lorentzian pulses, leaving only an ultranarrow, resonance-centered feature. Once the Fourier width has fallen below the coherence scale, increasing L further cannot continue to narrow this feature; its width is expected to approach the T_2-limited value $2/T_2$ in angular-frequency units from Eq. (S10). The present q9 archive does not contain an independent T_2 characterization, so Fig. S12 demonstrates convergence and ultranarrowing but does not by itself establish a quantitative equality to the T_2 limit.

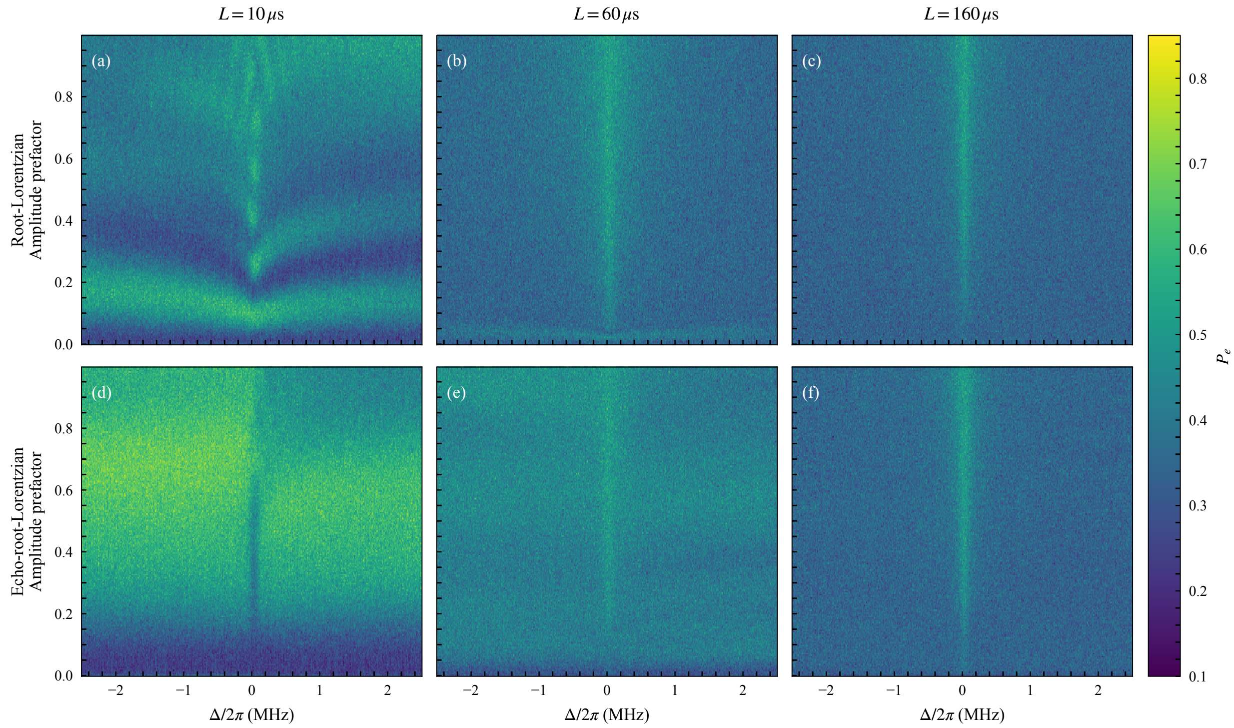

Figure S12. Experimental pulse-duration comparison on qubit q9. The top row shows root-Lorentzian spectroscopy and the bottom row shows the matched echo-root-Lorentzian protocol for $L = 10$, 60, and 160 μs. Each panel reports the measured excited-state probability over the same detuning and amplitude-prefactor grid; all panels share one color scale. The short root-Lorentzian scan exhibits pronounced amplitude-dependent structure, whereas the echo data are smoother. By $L = 160$ μs, both protocols have converged toward essentially the same response: a narrow resonance-centered line on a common relaxation-dominated background.

S9.2. Dependence on Pulse Cutoff

At fixed duration and peak Rabi frequency, varying the relative cutoff tests the sensitivity of both protocols to truncation of the Lorentzian tail. The relevant scale is the residual endpoint drive, $c\Omega_0$, which must remain small compared with the desired linewidth scale.

S9.3. Dependence on Drive Amplitude

Finally, amplitude sweeps compare the linewidth and contrast of the two pulse shapes over their full useful operating ranges. The echo protocol is assessed by the continuity and uniformity of its narrow-linewidth region, in addition to its minimum attainable linewidth.

S10. COMPARISON BETWEEN SIMULATION AND EXPERIMENT

Simulation and experiment are compared using the same final-state spectroscopic observable and the same linewidth-extraction procedure. The archived curves in the present figures use the historical coherence inputs described in Sec. S7, rather than the independently measured q1 values. Their agreement with the data is therefore interpreted qualitatively as support for the echo-cancellation mechanism, not as a parameter-matched validation. A quantitative comparison requires regenerating the simulations with $T_1 = 51.24 \pm 0.73$ μs and either the conservative Ramsey reference $T_{2,\text{eff}} = 7.31$ μs or a future measured Hahn-echo T_2. Systematic deviations at the largest amplitudes can also reflect physics omitted from the effective model, including higher-level coupling and strong-drive calibration effects.

S11. SIMULATED DURATION AND CUTOFF COMPARISON

Figures S13– S15 compare two-dimensional simulated spectroscopy maps for matched echo-root-Lorentzian and root-Lorentzian pulses. Each figure fixes the total duration at $L = 5$, 20, or 60 μs; the columns use relative endpoint cutoffs $c = 0.010$, 0.007, and 0.002. The upper row is the echo protocol and the lower row is the corresponding pulse without the central phase inversion. All panels use the same detuning, peak-Rabi-frequency, and color scales so that changes between protocols and cutoffs can be compared directly.

These preliminary maps are generated with the two-level Lindblad model of Sec. S7. They use the q1 measurement-linked values $T_1 = 51.24$ μs and $T_\phi = 7.87$ μs, corresponding to $T_{2,\text{eff}} = 7.31$ μs. Here $T_{2,\text{eff}}$ is the measured Ramsey T_2^* used as a conservative transverse time; it is not a measured Hahn-echo T_2. The maps span $\Delta/2\pi = -1$ to 1 MHz and $\Omega_0/2\pi = 0$ to 50 MHz. The generalized-Lorentzian exponent is $n = 1/2$, and the endpoint cutoff is imposed at fixed total duration using Eq. (S2). The nonuniform integration grid resolves the narrow pulse center and contains 1600 fourth-order Runge–Kutta steps per pulse half.

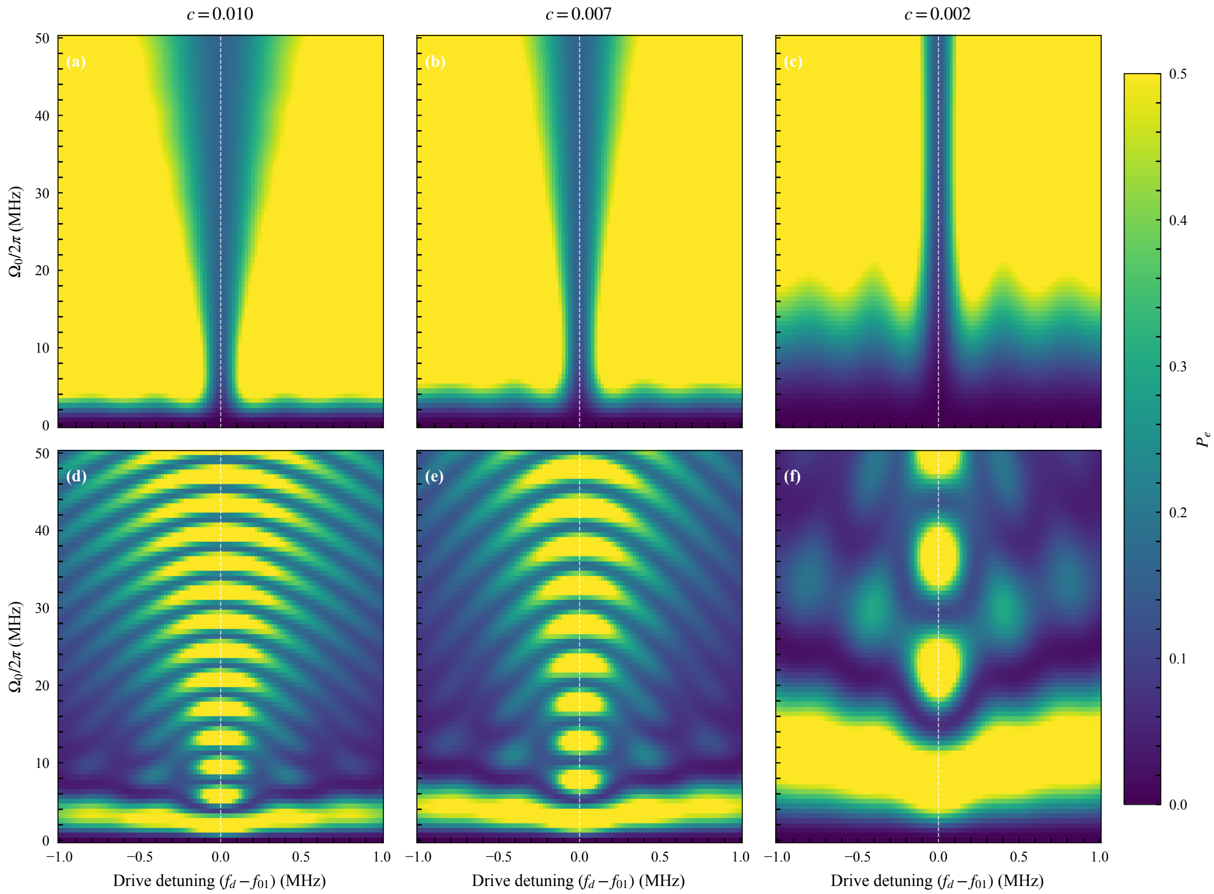

Figure S13. Simulated cutoff comparison for $L = 5$ μs. The upper row shows echo-root-Lorentzian spectroscopy and the lower row shows the matched root-Lorentzian protocol. Columns correspond to $c = 0.010$, 0.007, and 0.002. Every panel displays final excited-state probability over the same detuning and peak-Rabi-frequency grid.

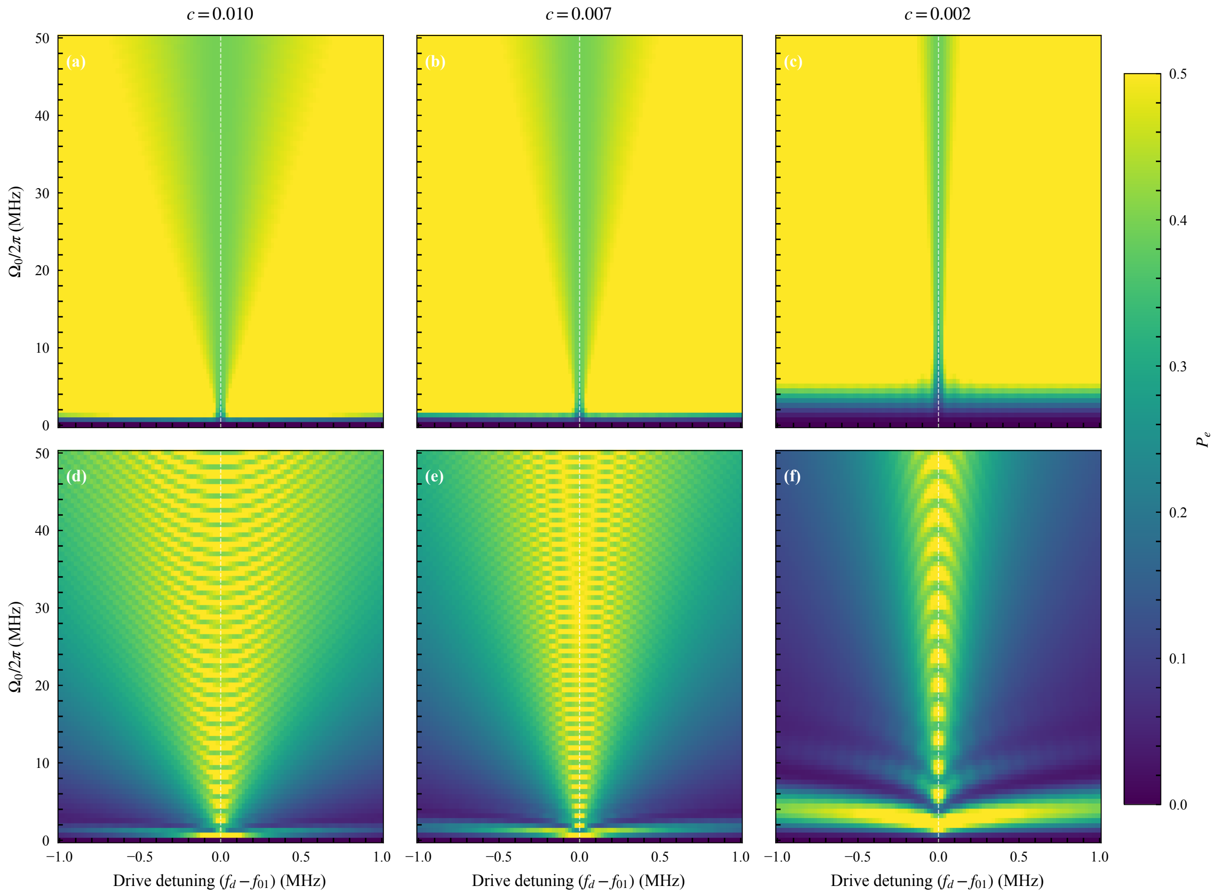

Figure S14. Simulated cutoff comparison for $L = 20\ \mu$s, using the same layout, axes, color scale, and model parameters as Fig. S13.

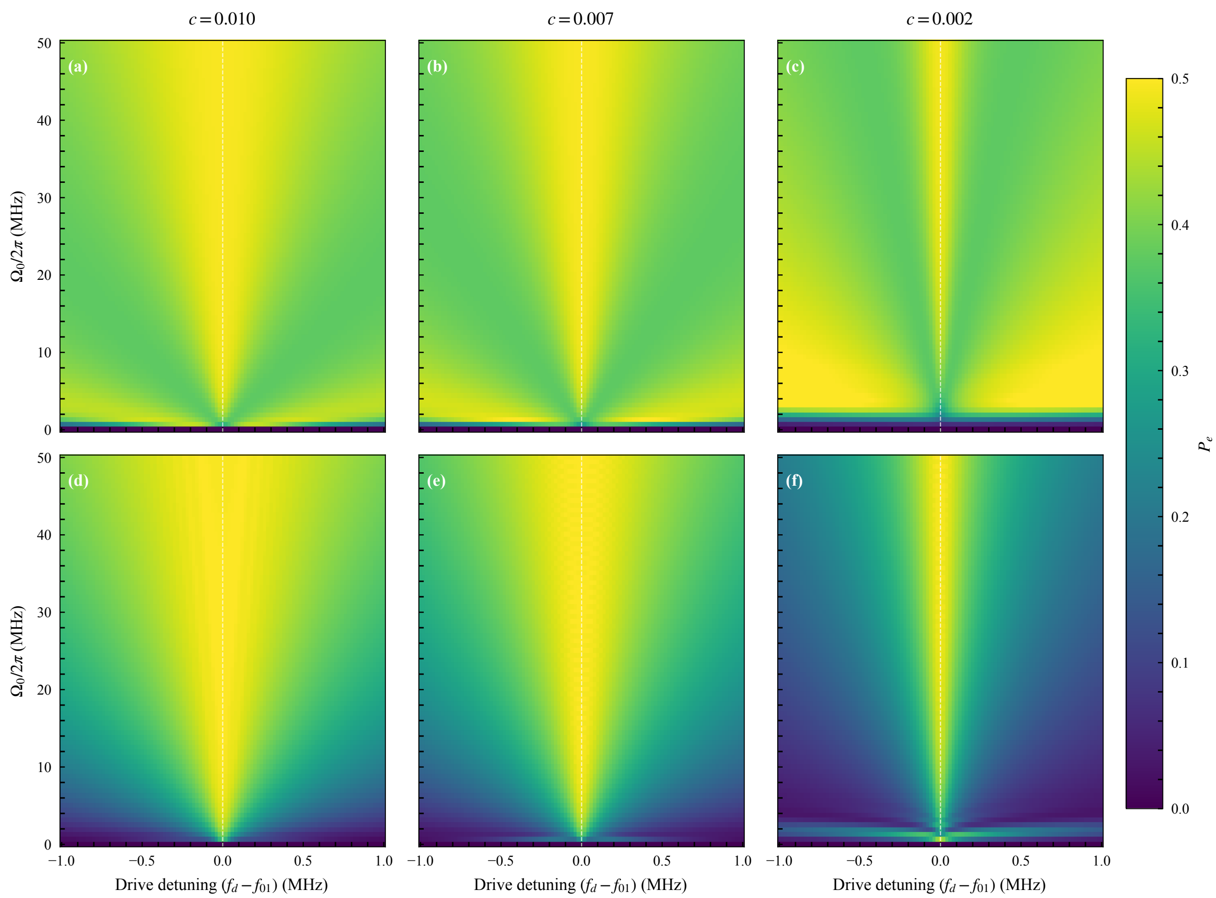

Figure S15. Simulated cutoff comparison for $L = 60\ \mu$s, using the same layout, axes, color scale, and model parameters as Fig. S13.

S12. THREE-LEVEL AC-STARK SHIFT OF THE f_{01} RESONANCE

We calculate the change of the central f_{01} resonance for a 40 μs constant square pulse and for 10 μs root-Lorentzian and echo-root-Lorentzian spectroscopy pulses. All calculations use the conventional drive–qubit detuning $\Delta = \omega_d - \omega_{01}$. The simulation retains $\{|g\rangle, |e\rangle, |f\rangle\}$, but $|f\rangle$ is used here as the virtual transmon level that shifts f_{01}; the plotted observable is the central excitation response $P_e + P_f$, not the position of the $g \leftrightarrow f$ transition. In this basis, the rotating-frame Hamiltonian of Eq. (S23) is

$$\frac{H(t)}{\hbar} = \begin{pmatrix} 0 & \Omega(t)/2 & 0 \\ \Omega(t)/2 & -\Delta & \Omega(t)/\sqrt{2} \\ 0 & \Omega(t)/\sqrt{2} & -2\Delta + \alpha \end{pmatrix}, \tag{S31}$$

where $\alpha = \omega_{12} - \omega_{01} < 0$. For a constant drive, virtual coupling between $|e\rangle$ and $|f\rangle$ gives the weak-drive result

$$\delta f_{01}^{(\mathrm{const})} \equiv \frac{\Delta_{\mathrm{res}}}{2\pi} \simeq -\frac{(\Omega_0/2\pi)^2}{2(\alpha/2\pi)}. \tag{S32}$$

Because $\alpha < 0$, the constant-pulse dressed f_{01} resonance moves to positive detuning. The exact values reported below are obtained by diagonalizing Eq. (S31) at constant $\Omega(t) = \Omega_0$ and minimizing the dressed g–e gap.

For the shaped pulses, $\Omega(t) = \Omega_0 f(t)$ with the root-Lorentzian envelope of Eq. (S1), order $n = 1/2$, and endpoint amplitude $c = 0.002$. A useful weak-drive phase-average estimate is

$$\delta f_{01}^{(\mathrm{shape})} \simeq -\frac{(\Omega_0/2\pi)^2}{2(\alpha/2\pi)} \left\langle f^2 \right\rangle_t, \qquad \left\langle f^2 \right\rangle_t = \frac{2\sigma}{L} \tan^{-1}\left(\frac{L}{2\sigma}\right), \tag{S33}$$

where $\sigma = (L/2)/\sqrt{c^{-2} - 1} = 0.010000$ μs and $\langle f^2 \rangle_t = 0.003138$. The echo phase inversion changes $f(t) \to -f(t)$ in the second half and therefore does not reverse this quadratic Stark term. Consequently, the root-Lorentzian and echo-root-Lorentzian pulses have the same leading phase-average estimate.

The experimentally visible feature of a finite pulse need not coincide with Eq. (S33), because the final population also contains coherent finite-pulse interference. We therefore report both quantities. For the ordinary root-Lorentzian pulse, the operational center is the local maximum of $P_e + P_f$ within $|\Delta|/(2\pi) \leq 0.5$ MHz. For the echo-root-Lorentzian pulse it is the corresponding central minimum. The spectra are evaluated on a 0.05 MHz grid and the extrema are refined with cubic interpolation. The central-feature extraction is shown for $\Omega_0/(2\pi) \geq 20$ MHz, where the feature has sufficient contrast.

The small negative position of the echo-root minimum at the largest drive is an operational finite-pulse feature position, not a negative instantaneous AC-Stark shift. It results from the interference defining the echo spectrum. The phase-average Stark term remains positive for both shaped protocols.

The time-domain maps solve the three-level Lindblad equation with $T_1 = 51.24$ μs, $T_\phi = 7.87$ μs, an initial $|g\rangle$ state, 31 peak Rabi frequencies from 0 to 60 MHz, and 101 detuning points over the full 100 MHz domain, from -50 to 50 MHz. Additional 401-point scans resolve the central 20 MHz span and the 1 MHz Lorentzian zooms in Figs. S16 and S17. The model uses the rotating-wave approximation, so it does not include a counter-rotating Bloch–Siegert shift.

Table S1. Calculated displacement of the f_{01} resonance or central finite-pulse feature, in MHz. The shaped phase average is common to the two Lorentzian-derived envelopes.

$\Omega_0/(2\pi)$	Constant dressed	Root maximum	Echo-root minimum	Shaped phase average
20	0.980	0.002	0.003	0.003
40	3.685	-0.001	-0.002	0.013
60	7.605	0.079	-0.013	0.028

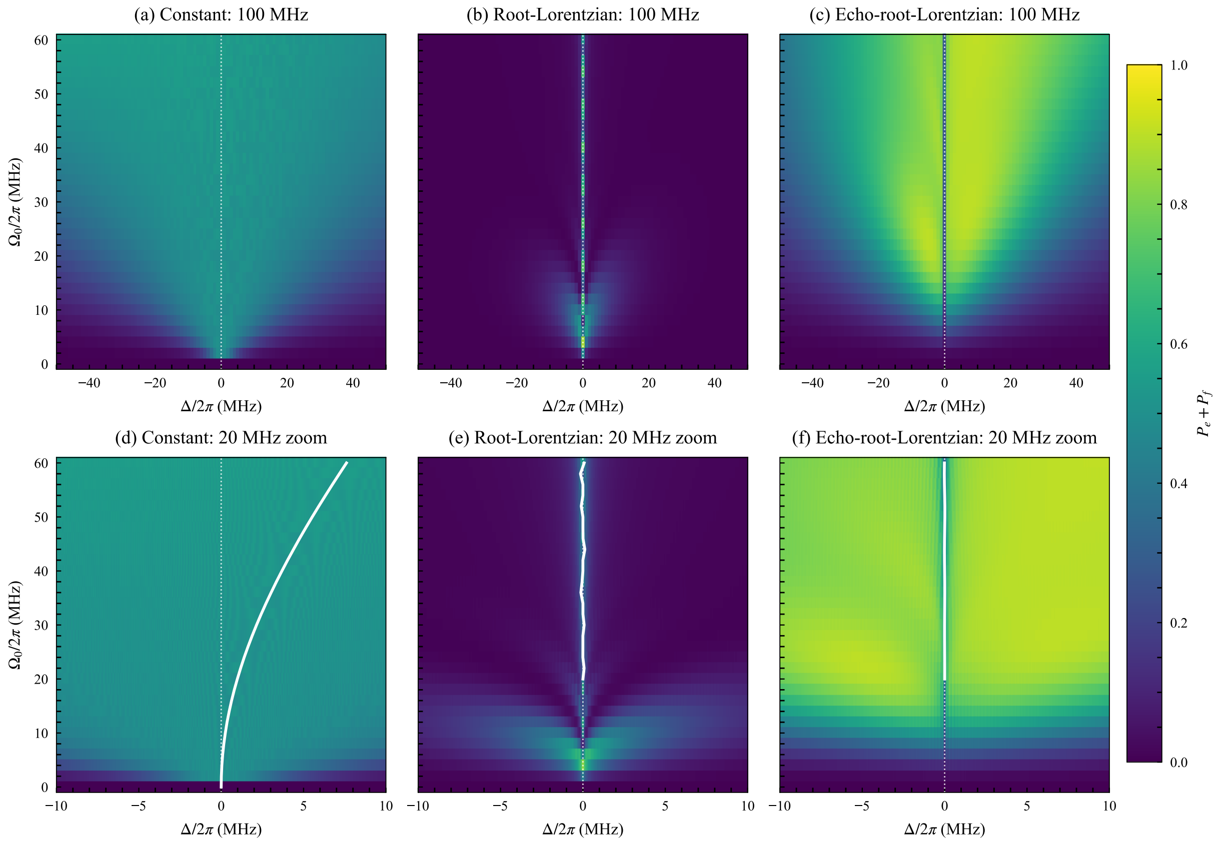

Figure S16. Full-domain and central-zoom three-level calculations of the f_{01} response for a 40 μs constant pulse and 10 μs shaped pulses. (a)–(c) Final excitation $P_e + P_f$ over the full -50 to 50 MHz domain for constant, root-Lorentzian, and echo-root-Lorentzian drives. (d)–(f) Corresponding 20 MHz zooms from -10 to 10 MHz. The dotted line is the bare f_{01} frequency. In the zooms, the solid lines show the exact constant-drive dressed center, the root-Lorentzian local maximum, and the echo-root-Lorentzian central minimum, respectively; shaped-pulse centers are omitted below 20 MHz peak Rabi frequency where the local-extremum definition is not robust.

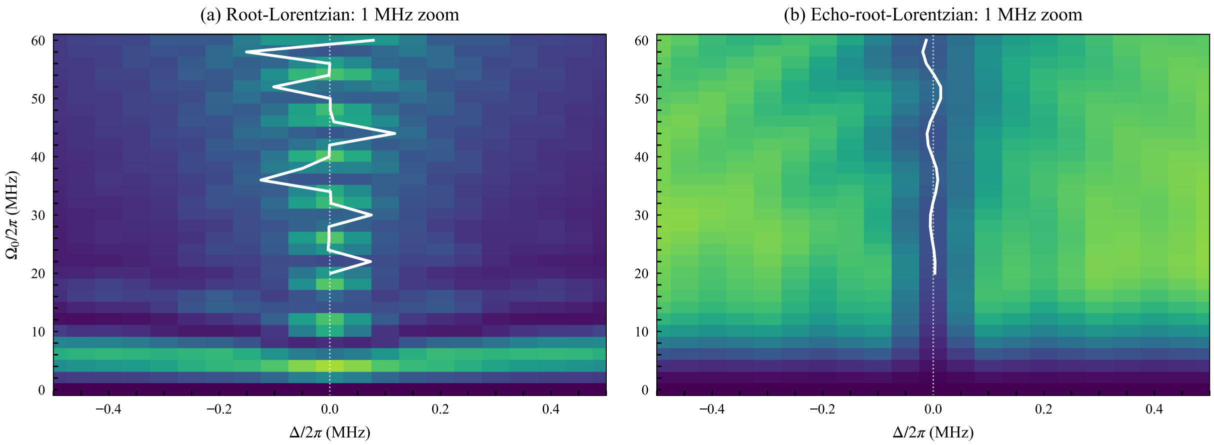

Figure S17. Fine f_{01} comparison for the Lorentzian-derived pulses. (a),(b) One-megahertz central zooms of the root-Lorentzian and echo-root-Lorentzian final excitation, respectively. The dotted line marks the bare f_{01} frequency and the solid lines mark the extracted maximum or central minimum.

S12.1. Measured resonance-center stability

The AC-Stark calculation above predicts a clear separation between the instantaneous shift produced by a constant drive and that produced by the Lorentzian-derived envelopes. At the upper edge of the experimentally selected operating window, $\Omega_0/(2\pi) = 8.973$ MHz, the weak-drive constant-pulse estimate from Eq. (S32) is approximately 0.20 MHz. Multiplication by the shaped-pulse factor $\langle f^2 \rangle_t = 0.003138$ reduces the phase-average estimate to 0.63 kHz. The wider-range three-level calculation in Figure 4 of the Letter confirms this separation over the wider calculated drive range. The echo phase inversion does not cancel the quadratic Stark term; rather, the small time-averaged squared envelope suppresses it, while finite-pulse interference determines the experimentally visible central minimum.

We test the corresponding center stability directly in the q1 amplitude-sweep data used in Figs. S8–S10. Each amplitude slice is fit with the same Gaussian dip estimator and preprocessing described in Sec. S8. To avoid selecting the result being tested, the center-quality mask uses only finite fit parameters, contrast $A \geq 0.02$, and FWHM $\Gamma_f \leq 0.12$ MHz; it does not impose the usual $|\delta_0| \leq 0.050$ MHz center gate. We then restrict the summary to the independently selected matched-simulation operating window $\Omega_0/(2\pi) = 2.475$–8.973 MHz, defined by $Q \geq 0.10$.

The inverse-variance weighted center is $\overline{\delta_0} = -0.47 \pm 0.15$ kHz, where the stated uncertainty uses the fit covariance only. Relative to this center, the 42 retained amplitude slices have an empirical RMS displacement of 3.02 kHz and a maximum absolute displacement of 7.47 kHz. The constant-fit value $\chi^2_\nu = 9.94$ shows that the slice-to-slice scatter is larger than the local covariance errors; the measurements therefore do not establish strict statistical constancy. Instead, the RMS and maximum excursion quantify the practical center stability, including fit variation, slow experimental drift, and any residual finite-pulse or AC-Stark contribution. The observed few-kilohertz, nonmonotonic excursions are nevertheless far smaller than the approximately 0.20 MHz constant-drive Stark displacement expected at the same upper peak Rabi frequency.

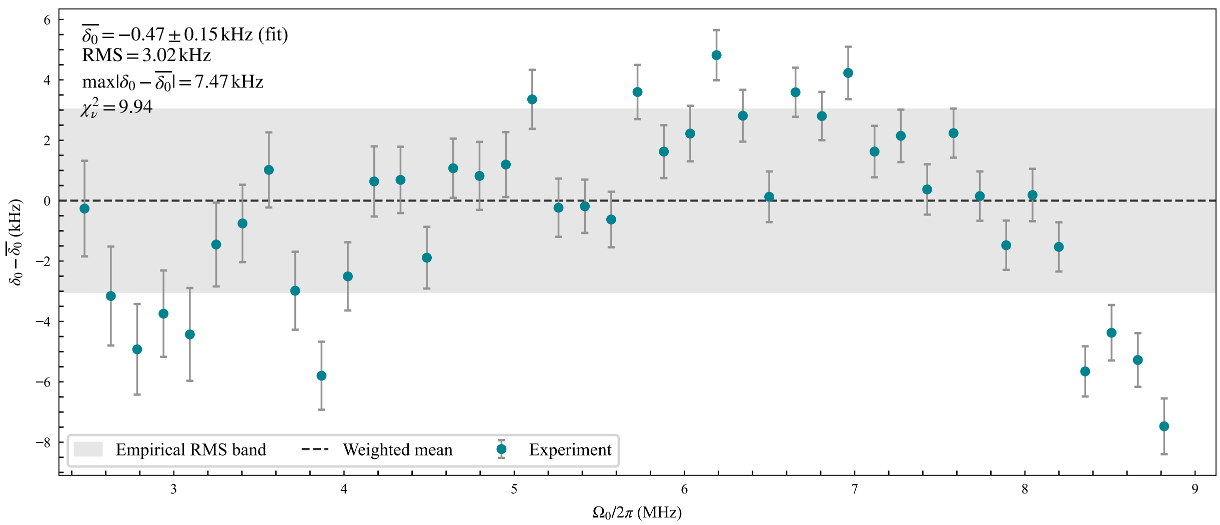

Figure S18. Measured resonance-center stability across the independently selected q1 echo–root–Lorentzian operating window. Points show the fitted center displacement relative to the inverse-variance weighted mean; vertical bars are one-standard-deviation covariance uncertainties. The dashed line is the weighted mean and the gray band spans the empirical RMS displacement of ± 3.02 kHz. The maximum absolute displacement is 7.47 kHz. No center-position gate is used to select the displayed points. Although $\chi^2_\nu = 9.94$ rules out interpreting the covariance errors as a complete noise model, the bounded center excursion directly quantifies amplitude robustness and is strongly separated from the constant-drive AC-Stark displacement in Fig. 4 of the Letter.

S12.2. Three-level optimization of resonance-center stability

We further test whether a derivative quadrature and a dynamic frequency correction can stabilize the central echo-root-Lorentzian feature in the three-level model. The reproducible calculation is contained in `notebooks/paper/34_drag_echo_lorentzian_qutrit.ipynb`. It uses a 10 μs pulse with $c = 0.002$, echo inversion enabled, $\alpha/(2\pi) = -200$ MHz, $T_1 = 51.24$ μs, and $T_\phi = 7.87$ μs. The final populations P_g, P_e, and P_f are calculated over peak Rabi frequencies from 0 to 80 MHz and detunings from -1 to 1 MHz.

The complex drive is written as $\Omega_I - i\Omega_Q$. Its derivative quadrature is

$$\Omega_Q(t) = -\frac{\beta}{\alpha}\frac{d\Omega_I(t)}{dt}, \tag{S34}$$

where the derivative acts on the smooth positive envelope within each half of the pulse and the same $0/\pi$ echo phase is applied to both quadratures. The ideal instantaneous phase jump itself is not differentiated. We use the moderate value $\beta = 2$, for which the peak quadrature is 6.11% of the peak in-phase drive. In addition, the instantaneous drive detuning is corrected according to

$$\frac{\Delta_{\mathrm{corr}}(t)}{2\pi} = \kappa \left[\frac{\Omega_I(t)}{2\pi}\right]^2, \tag{S35}$$

with κ expressed in MHz^{-1}. This is equivalent to a phase chirp satisfying $\dot{\phi}(t) = -\Delta_{\mathrm{corr}}(t)$, up to the stated detuning convention.

For each candidate κ, the notebook calculates a compact map over $\Omega_0/(2\pi) = 20$–80 MHz. The central minimum of P_e is identified within $|\Delta|/(2\pi) \leq 0.35$ MHz and refined with a local quadratic fit. The optimization target is its RMS displacement from the bare transition,

$$\mathcal{L} = \sqrt{\langle \delta_0^2(\Omega_0) \rangle_{\Omega_0}}, \tag{S36}$$

while the maximum P_f is retained as an independent leakage diagnostic. A coarse scan over $\kappa = -0.005$ to 0.005 MHz^{-1} is followed by a fine scan around the best coarse value. As shown in Fig. S19(a), the optimum for this model is

$$\kappa = +0.001750 \ \mathrm{MHz}^{-1}. \tag{S37}$$

The fitted central minimum then has an RMS displacement of 3.54 kHz from the bare transition and a maximum absolute displacement of 6.60 kHz over the optimization amplitudes. These values are simulation results, not experimentally calibrated correction parameters.

The full-resolution comparison is shown in Fig. S19(b). Relative to the uncorrected echo-root-Lorentzian calculation, the combined DRAG and Stark-corrected pulse reduces the maximum P_f from 0.1781 to 0.1598. Over $\Omega_0/(2\pi) = 40$–80 MHz, the mean P_f decreases from 0.0669 to 0.0614. The low-amplitude population response retains some coherent oscillatory structure: the optimization in Eq. (S36) constrains the resonance position and does not explicitly minimize population ripple. Thus the result demonstrates improved center stability and leakage within the model, but not complete removal of all finite-pulse oscillations.

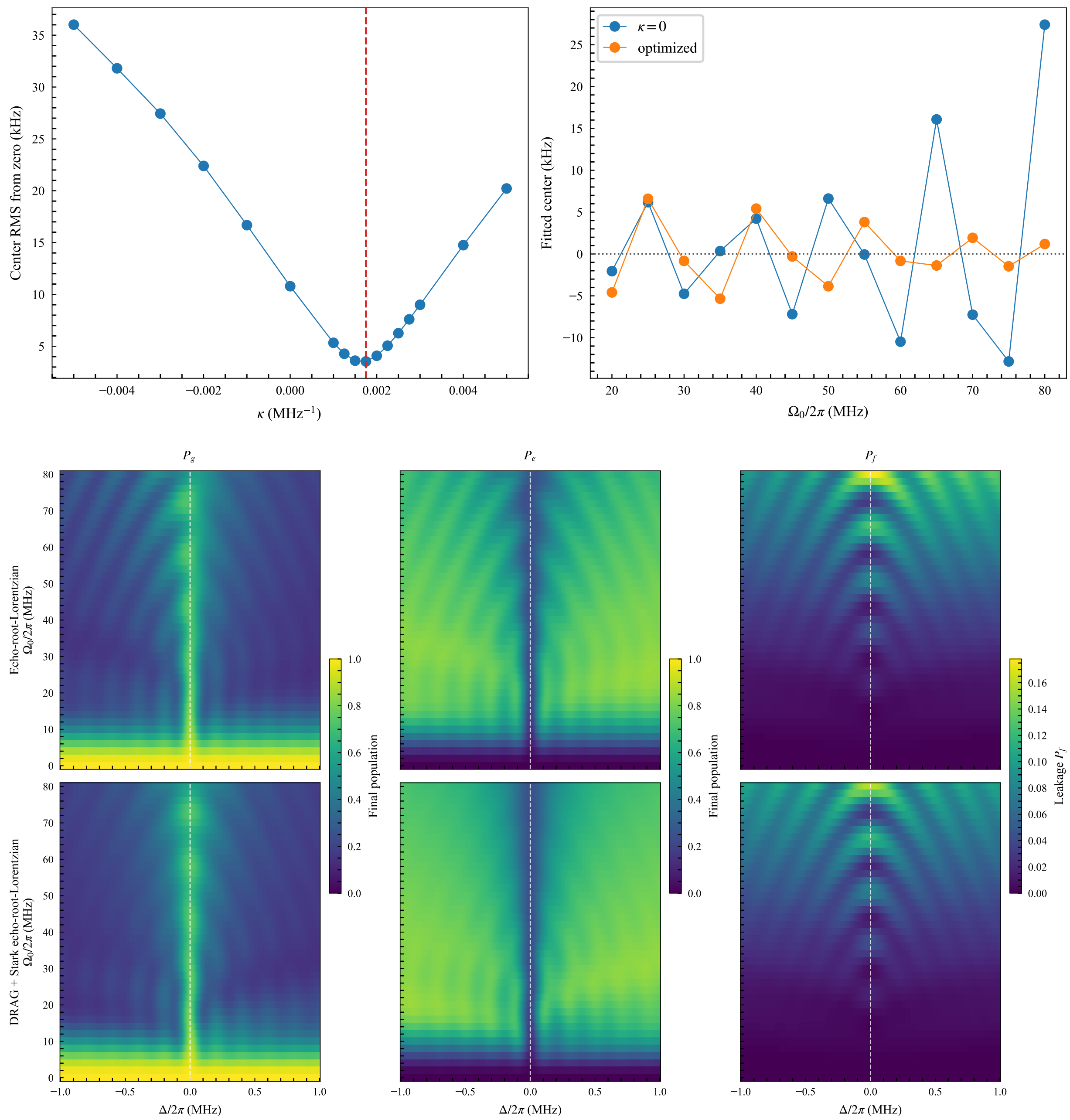

Figure S19. Three-level optimization of the echo-root-Lorentzian response. (a) Quadratic Stark-correction scan. The left panel shows the RMS fitted-center displacement from the bare transition versus κ; the dashed line marks the selected coefficient. The right panel compares the fitted central minimum without the correction and at the optimized value. (b) Full -1 to 1 MHz detuning maps. The upper row is the original echo pulse and the lower row includes the $\beta = 2$ derivative quadrature and optimized quadratic Stark correction. Columns show P_g, P_e, and P_f; dashed lines mark the bare transition.

[1] Quantum Machines, OPX1000 front end modules: Microwave FEM, `https://docs.quantum-machines.co/latest/docs/Guides/opx1000_fems/` (2026), accessed 19 July 2026.

[2] Quantum Machines, Demodulation and measurement, `https://docs.quantum-machines.co/latest/docs/Guides/demod/` (2026), accessed 19 July 2026.

[3] F. Bloch, Nuclear induction, Phys. Rev. **70**, 460 (1946).

[4] I. S. Mihov and N. V. Vitanov, Defying conventional wisdom in spectroscopy: Power narrowing on IBM quantum, Physical Review Letters **132**, 020802 (2024), arXiv:2308.14187 [quant-ph].

[5] I. I. Boradjiev and N. V. Vitanov, Power narrowing in coherent atomic excitation by smoothly shaped pulsed fields, Optics Communications **288**, 91 (2013).